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Discovering a radio controller in a cloud radio access network

Abstract

A distributed radio access network (RAN) includes a new radio unit (RU) being connected to a front-haul network. The new RU determines its own Internet Protocol (IP) address based on its MAC address; and send a discovery message to all radio controllers in the front-haul network via a multicast address. The distributed RAN also includes a plurality of radio controllers communicatively coupled to the new RU via the front-haul network. Each of the plurality of radio controllers are configured to receive the discovery message and determine whether the new RU is assigned to the respective radio controller. Each of at least one radio controller, to which the new RU is assigned, establishes at least one configuration session with the new RU.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0088610	12/2015	Abraham et al.	N/A	N/A
2017/0257150	12/2016	Chae et al.	N/A	N/A
2018/0191873	12/2017	Bao et al.	N/A	N/A
2019/0245740	12/2018	Kachhla	N/A	N/A
2019/0320494	12/2018	Jayawardene	N/A	H04L 61/5007
2020/0092171	12/2019	Ramakrishnan	N/A	H04L 41/0809
2020/0204252	12/2019	Barbieri et al.	N/A	N/A

OTHER PUBLICATIONS

International Searching Authority, “International Search Report and Written Opinion” from PCT Application No. PCT/US2022/016146 dated May 26, 2022, from Foreign Counterpart to U.S. Appl. No. 17/670,022, pp. 1 through 11, Published in: KR. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/157,961 filed on Mar. 8, 2021, entitled “DISCOVERING A RADIO CONTROLLER IN A CLOUD RADIO ACCESS NETWORK”, the entirety of which is incorporated herein by reference.

BACKGROUND

(1) In a distributed radio access network (RAN), geographically separated radio units (RUs) are controlled by centralized unit(s) and provide wireless service to nearby user equipment (UEs). It is desirable for an RU to dynamically discover a radio controller upon being powered up.

SUMMARY
(2) A distributed radio access network (RAN) includes a new radio unit (RU) being connected to a front-haul network. The new RU determines its own Internet Protocol (IP) address based on its MAC address; and send a discovery message to all radio controllers in the front-haul network via a multicast address. The C-RAN also includes a plurality of radio controllers communicatively coupled to the new RU via the front-haul network. Each of the plurality of radio controllers are configured to receive the discovery message and determine whether the new RU is assigned to the respective radio controller. Each of at least one radio controller, to which the new RU is assigned, establishes at least one configuration session with the new RU.
(3) A distributed radio access network (RAN) includes a new radio unit (RU) being connected to a plurality of radio controllers via a front-haul network. The new RU is also configured to request, from a Dynamic Host Configuration Protocol (DHCP) server after the new RU powers up a first time, IP addresses for each radio controller it is assigned to. The DHCP server is configured to send a response to the new RU indicating a number of IP addresses equal to a number of carriers the new RU needs to establish, each of the IP addresses belonging to a radio controller the new RU is assigned to. The new RU initiates a different configuration session for each carrier it needs to establish with each radio controller it is assigned to.

Description

DRAWINGS
(1) Understanding that the drawings depict only exemplary configurations and are not therefore to be considered limiting in scope, the exemplary configurations will be described with additional specificity and detail through the use of the accompanying drawings, in which:
(2) FIG. **1**A is a block diagram illustrating an exemplary configuration of a system implementing radio controller discovery in a C-RAN;
(3) FIG. **1**B is a block diagram illustrating another exemplary configuration of a system implementing radio controller discovery in a C-RAN;
(4) FIG. **2**A is a block diagram illustrating a multi-operator system with multi-instance RUs;
(5) FIG. **2**B is a block diagram illustrating another multi-gNB system with multi-instance RUs;
(6) FIG. **3**A is a block diagram illustrating the first example configuration of radio controller discovery in a system;
(7) FIG. **3**B is a block diagram illustrating different portions of an IPv6 address and how an IPv6 address can be derived from a MAC address;
(8) FIG. **4** is a flow diagram illustrating a method for radio controller discovery according to the first example configuration described herein;
(9) FIG. **5** is a block diagram illustrating the second example configuration of radio controller discovery in a system; and
(10) FIG. **6** is a flow diagram illustrating a method for radio controller discovery according to the second example configuration described herein.
(11) In accordance with common practice, the various described features are not drawn to scale but are drawn to emphasize specific features relevant to the exemplary configurations.
DETAILED DESCRIPTION
(12) In some configurations, a Third Generation Partnership Project (3GPP) Fourth Generation (4G) C-RAN will include multiple devices (e.g., baseband controller unit(s), radio units (RUs), RF units (antenna arrays)) interfaced together to enable and implement RAN functions. In a typical 3GPP Fifth Generation (5G) C-RAN, there may be further logical and/or physical splits in the devices (e.g., centralized unit(s) (CU(s)), distributed unit(s) (DU(s)), radio units (RUs), RF units (antenna arrays)), which are interfaced together to enable and implement RAN functions.

(13) The Open Radio Network Alliance (O-RAN) Alliance is seeking to standardize certain aspects of cloud radio access network operation. In O-RAN specifications (available at https://www.o-ran.org/specifications), RUs are controlled by a “radio controller”, which can be implemented in a baseband controller (in 4G) or a DU or CU (in 5G). In the examples below, the radio controller is generally described as being located in the DU(s), however other configurations are possible. Thus, where the term “radio controller” is used herein, it may refer to a 4G baseband controller, a 5G DU, or a 5G CU.
(14) A distributed antenna system (DAS) (not shown) is another type of distributed RAN that includes at least two RUs **106** and a centralized distribution unit (sometimes referred to as a “head unit”). The RUs **106** can wirelessly transmit signals to UEs **110** in a coverage area. The distribution unit can communicate channelized digital baseband signals with the RUs **106**. The channelized digital baseband signals may include call information for wireless communication. The DAS may implement additional devices and/or functionality. A DAS may implement any suitable air interface, e.g., Third Generation Partnership Project (3GPP) 3G, 4G, and/or 5G air interface(s). In some configurations, a radio controller can be implemented in a head unit in a distributed antenna system. Thus, while C-RANs are generally used in the examples below, the present systems and methods could equally apply to different implementations of distributed RANs, e.g., DAS.
(15) Upon joining a front-haul network (e.g., powering up for the first time after being connected to the front-haul network) in a C-RAN, a RU needs to be assigned an IP address and discover its radio controller (that will configure and manage the RU). The O-RAN specification proposes different ways for the RU to be allocated an IP address and discover its radio controller. One proposed way is to have that IP address of the radio controller and/or RU pre-shared (e.g., hard-coded) on both radio controllers and RUs so they can communicate with each other upon power up without any discovery procedures.
(16) A more dynamic O-RAN proposal is for the RU to contact a Dynamic Host Configuration Protocol (DHCP) server after power up, after which the DHCP server assigns the RU an IP address and also sends the IP address of its radio controller to the RU. While IPv6 IP addresses are discussed in the examples below, it should be noted that any of the IP addresses may alternatively be Internet Protocol version 4 (IPv4) IP addresses.
(17) In contrast, in a first example configuration, the present systems and methods use a dynamic approach for radio controller discovery that does not require a DHCP server. Specifically, in the first example configuration, a radio unit (RU) determines its IP address (e.g., its IPv6 address) and multicasts a neighbor solicitation (NS) message to all devices on the same network. In response, only the radio controller(s) (e.g., 5G DU(s)) to which the RU is assigned will respond with a neighbor advertisement and initiate a Network Configuration Protocol (NETCONF) session with the RU to configure the RU for operation. Each radio controller may be pre-loaded with a whitelist that indicates MAC addresses (e.g., which can also serve as the serial numbers) of RUs that are assigned to it, so each radio controller can (1) determine whether it needs to respond to a neighbor solicitation from a particular RU (e.g., based on whether the RU's Media Access Control (MAC) address is listed in the radio controller's whitelist); and (2) if the RU is assigned to the radio controller, the radio controller may initiate a NETCONF session with the RU (e.g., using the RU's link-local IPv6 address from the configured MAC address) without waiting for the RU to initiate the NETCONF session. Because the radio controller initiates a NETCONF session using the whitelist, traffic is reduced within the front-haul network of a C-RAN compared to other non-DHCP solutions. For example, if all radio controllers were instead required to respond to every RU's neighbor solicitation in a C-RAN (e.g., with multiple DUs and up to 128 RUs), the resulting traffic from excess neighbor acknowledgements may unnecessarily consume bandwidth in the front-haul network.
(18) While NETCONF is used in examples herein, any suitable configuration management protocol can be used between the radio controller and RU in order to configure the RU.

(19) In a second example configuration, the present systems and methods extend the O-RAN DHCP proposal, which limits communication to one radio controller and one RU, i.e., in O-RAN the DHCP server sends back to the RU a single radio controller Internet Protocol (IP) address and a single RU IP address. However, RUs can, in some cases, initiate multiple connections with multiple radio controllers (e.g., DUs). For example, in some configurations, RUs can include N-carrier hardware (e.g., where N=4) that is capable of simultaneously working with up to four 5G gNB DUs and/or 4G LTE baseband controllers to serve up to 4 different cells.
(20) Accordingly, in the second example configuration of the present systems and methods, the DHCP server sends back IP addresses for one or more radio controllers (and if the radio controller can initiate N connections with the same RU, its IP address will be listed N times). So when the RU gets this list from the DHCP server, it knows how many connections to each radio controller it needs to entertain. Thus, the second example configuration herein allows discovery of radio controllers in a multi-carrier and/or multi-operator system, which is not contemplated by the O-RAN DHCP proposal.
(21) Example 5G C-RAN
(22) FIG. **1**A is a block diagram illustrating an exemplary configuration of a system **101**A implementing radio controller discovery in a C-RAN **100**A. In the exemplary embodiment shown in FIG. **1**A, a base station employs a centralized or cloud RAN (C-RAN) architecture for each cell (or sector) **102** served by the base station **100**A, with the following logical nodes: at least one control unit (CU) **103**, at least one distributed unit (DU) **105**, and multiple radio units (RUs) **106**. Each RU **106** is remotely located from each CU **103** and DU **105** serving it. Also, in this exemplary embodiment, at least one of the RUs **106** is remotely located from at least one other RU **106** serving that cell **102**.
(23) The C-RAN **100**A can be implemented in accordance with one or more public standards and specifications. In some configurations, the C-RAN **100**A is implemented using the logical RAN nodes, functional splits, and front-haul interfaces defined by the O-RAN Alliance. In such an O-RAN example, each CU **103**, DU **105**, and RU **106** can be implemented as an O-RAN central unit (CU), O-RAN distributed unit (DU), and O-RAN radio unit (RU), respectively, in accordance with the O-RAN specifications.
(24) That is, each CU **103** comprises a logical node hosting Packet Data Convergence Protocol (PDCP), Radio Resource Control (RRC), Service Data Adaptation Protocol (SDAP), and other control functions. Therefore, each CU **103** implements the gNB controller functions such as the transfer of user data, mobility control, radio access network sharing, positioning, session management, etc. The CU(s) **103** control the operation of the Distributed Units (DUs) **105** over an interface (including F1-c and F1-u for the control plane and user plane, respectively).
(25) In FIG. **1**A, the C-RAN **100**A includes a single CU **103**, which handles control plane functions, user plane functions, some non-real-time functions, and/or Packet Data Convergence Protocol (PDCP) processing. The CU **103** (in the C-RAN **100**A) may communicate with at least one wireless service provider's Next Generation Cores (NGC) **112** using 5G NGc and 5G NGu interfaces. In some 5G configurations (not shown in FIG. **1**A), a CU **103** is split between a CU-CP that handles control plane functions and a CU-UP that handles user plane functions.
(26) In some configurations, each DU **105** comprises a logical node hosting (performing processing for) Radio Link Control (RLC) and Media Access Control (MAC) layers, as well as optionally the upper or higher portion of the Physical (PHY) layer (where the PHY layer is split between the DU **105** and RU **106**). In other words, the DUs **105** implement a subset of the gNB functions, depending on the functional split (between CU **103** and DU **105**). In some configurations, the L3 processing (of the 5G air interface) may be implemented in the CU **103** and the L2 processing (of the 5G air interface) may be implemented in the DU **105**. As noted above, a DU **105** (or a CU **103**) may act as a “radio controller” for one or more RUs **106** in a 5G C-RAN **100**A operating according to O-RAN specifications.

(27) Each RU **106** comprises a logical node hosting the portion of the PHY layer not implemented in the DU **105** (that is, the lower portion of the PHY layer) as well as implementing the basic RF and antenna functions. In some 5G configurations, the RUs (RUs) **106** may communicate baseband signal data to the DUs **105** on an NG-iq interface. In some 5G configurations, the RUs **106** may implement at least some of the L1 and/or L2 processing. In some configurations, the RUs **106** may have multiple ETHERNET ports and can communicate with multiple switches.
(28) Although the CU **103**, DU **105**, and RUs **106** are described as separate logical entities, one or more of them can be implemented together using shared physical hardware and/or software. For example, in the exemplary embodiment shown in FIG. **1**A, for each cell **102**, the CU **103** and DU **105** serving that cell **102** could be physically implemented together using shared hardware and/or software, whereas each RU **106** would be physically implemented using separate hardware and/or software. Alternatively, the CU(s) **103** may be remotely located from the DU(s) **105**.
(29) Also, in the exemplary embodiment described here in connection with FIG. **1**A, the C-RAN **100**A is implemented as a Fifth Generation New Radio (5G NR) RAN that supports a 5G NR wireless interface in accordance with the 5G NR specifications and protocols promulgated by the 3rd Generation Partnership Project (3GPP). Thus, in some configurations, the C-RAN **100**A can also be referred to as a “Next Generation Node B” **100**, “gNodeB” **100**, or “gNB” **100**.
(30) Each RU **106** includes or is coupled to one or more antennas **122** via which downlink RF signals are radiated to various items of user equipment (UE) and via which uplink RF signals transmitted by UEs **110** are received.
(31) The CU **103** and/or DU(s) **105** may be coupled to a core network **112** of the associated wireless network operator over an appropriate back-haul network **116** (such as the Internet). Also, each DU **105** is communicatively coupled to the RUs **106** served by it using a front-haul network **118**. Each of the DU(s) **105** and RUs **106** include one or more network interfaces (not shown) to enable the DU(s) **105** and RUs **106** to communicate over the front-haul network **118**.
(32) In one implementation, the front-haul **118** that communicatively couples the DU(s) **105** to the RUs **106** is implemented using a switched ETHERNET network **120**. In such an implementation, each DU **105** and RU **106** includes one or more ETHERNET interfaces for communicating over the switched ETHERNET network **120** used for the front-haul **118**. However, it is to be understood that the front-haul **118** between each DU **105** and the RUs **106** served by it can be implemented in other ways.
(33) Each CU **103**, DU **105**, and RU **106**, (and the functionality described as being included therein), as well as any other device in the system **101**A more generally, and any of the specific features described here as being implemented by any of the foregoing, can be implemented in hardware, software, or combinations of hardware and software, and the various implementations (whether hardware, software, or combinations of hardware and software) can also be referred to generally as “circuitry” or a “circuit” or “circuits” configured to implement at least some of the associated functionality. When implemented in software, such software can be implemented in software or firmware executing on one or more suitable programmable processors or configuring a programmable device (for example, processors or devices included in or used to implement special-purpose hardware, general-purpose hardware, and/or a virtual platform). Such hardware or software (or portions thereof) can be implemented in other ways (for example, in an application specific integrated circuit (ASIC), etc.). Also, the RF functionality can be implemented using one or more RF integrated circuits (RFICs) and/or discrete components. Each CU **103**, DU **105**, RU **106**, and the system **101**A more generally, can be implemented in other ways.
(34) As noted above, in the exemplary configuration described here in connection with FIG. **1**A, the C-RAN **100**A is implemented as a 5G NR RAN that supports a 5G NR wireless interface to wirelessly communicate with the UEs **110**.
(35) More specifically, in the exemplary embodiment described here in connection with FIG. **1**A, the 5G NR wireless interface may support the use of beamforming for wirelessly communicating

with the UEs **110** in both the downlink and uplink directions using the millimeter wave (mmWave) radio frequency (RF) range defined for 5G NR (Frequency Range 2 or “FR2”), e.g., ranging from 24 GHz to 40 or 100 GHz. 5G NR RAN systems typically make use of fine beams and beamforming, especially when FR2 is used. To perform such beamforming, each RU **106** comprises an array of multiple, spatially separated antennas **122**. When FR2 is used, the spacing of the antennas **122** in the array is on the order of several millimeters (as opposed to several centimeters as is the case when FR1 is used) and can be implemented in a convenient fashion.
(36) In some configurations, the C-RAN **100**A may implement uplink combining in which a group of RUs **106** (e.g., up to four) receive RF signals from a particular UE **110** and a DU **105** and/or CU **103** combines them (e.g., using a maximum likelihood combining) into a single uplink signal. Additionally or alternatively, the C-RAN **100**A may implement downlink combining in which a group of RUs **106** send downlink RF signals to a particular UE **110**, which combines them (e.g., using a maximum likelihood combining) into a single downlink signal.
(37) A management system **114** may be communicatively coupled to the CU(s) **103**, DU(s) **105**, and/or RUs **106**, for example, via the back-haul network **116** and/or the front-haul network **118**. The management system **114** may assist in managing and/or configuring the C-RAN **100**A. A hierarchical architecture can be used for management-plane (“M-plane”) communications. When a hierarchical architecture is used, the management system **114** can send and receive M-plane (management) communications to and from the DU **105**, which in turn forwards relevant M-plane communications to and from the RUs **106** as needed. Alternatively, a direct architecture can also be used for M-plane communications. When a direct architecture is used, the management system **114** can communicate directly with the RUs **106** (without having the M-plane communications forwarded by the CU **103** or DU **105**). A hybrid architecture can also be used in which some M-plane communications are communicated using a hierarchical architecture and some M-plane communications are communicated using a direct architecture. Proprietary protocols and interfaces can be used for such M-plane communications. Also, protocols and interfaces that are specified by standards such as O-RAN can be used for such M-plane communications.
(38) Example 4G C-RAN
(39) FIG. **1**B is a block diagram illustrating another exemplary configuration of a communication system **101**B implementing radio controller discovery in a C-RAN **100**B. In the exemplary configuration shown in FIG. **1**B, the system **101**B is implemented using a cloud radio access network (C-RAN) (point-to-multipoint distributed base station) architecture that employs at least one baseband unit **104** and one or more radio units **106**.
(40) The RUs **106** may be deployed at a site **115** to provide wireless coverage and capacity for one or more wireless network operators. The site **115** may be, for example, a building or campus or other grouping of buildings (used, for example, by one or more businesses, governments, other enterprise entities) or some other public venue (such as a hotel, resort, amusement park, hospital, shopping center, airport, university campus, arena, or an outdoor area such as a ski area, stadium, or a densely-populated downtown area). In some configurations, the site **115** is at least partially (and optionally entirely) indoors, but other alternatives are possible.
(41) The C-RAN **100**B may include a baseband unit **104**, which may also be referred to as “baseband controller” **104**, or just “controller” **104**. Each radio unit (RU) **106** may include or be coupled to at least one antenna used to radiate downlink RF signals to user equipment (UEs) **110** and receive uplink RF signals transmitted by UEs **110**. The baseband controller **104** may optionally be physically located remotely from the site **115**, e.g., in a centralized bank of baseband controllers **104**. Additionally, the RUs **106** may be physically separated from each other within the site **115**, although they are each communicatively coupled to the baseband controller **104** via a front-haul network **118** (or just “front-haul”). Communication relating to L1 functions generally relies on the front-haul network **118** interface. As before, every RU **106** in the system **100**B may transmit the same or different cell-ID for each of the cell(s) they all serve, depending on the number of carriers

and frequency reuse layers. As noted above, a baseband controller **104** may be referred to as a "radio controller" for one or more RUs **106** in a 4G C-RAN **100**B operating according to O-RAN specifications.

(42) Each UE **110** may be a computing device with at least one processor that executes instructions stored in memory, e.g., a mobile phone, tablet computer, mobile media device, mobile gaming device, laptop computer, vehicle-based computer, a desktop computer, etc. Each baseband controller **104** and RU **106** may be a computing device with at least one processor that executes instructions stored in memory. Furthermore, each RU **106** may optionally implement one or more RU instances, e.g., a processing core that executes instructions that implement the functionality of an RU **106**.

(43) The C-RAN **100**B may optionally implement frequency reuse where the same frequency resource(s) are used for multiple sets of UEs **110**, each set of UEs **110** being under a different, geographically diverse set of RUs **106** (all serving the same cell).

(44) The system **100**B is coupled to a core network **112** of each wireless network operator over an appropriate back-haul network **116**. For example, the Internet may be used for back-haul **116** between the system **100**B and each core network **112**. However, it is understood that the back-haul network **116** can be implemented in other ways. Communication relating to L3 functions generally relies on the back-haul network **116** interface. Each of the back-haul network **116** and/or the front-haul network **118** described herein may be implemented with one or more network elements, such as switches, routers, and/or other networking devices. For example, the back-haul network **116** and/or the front-haul network **118** may be implemented as a switched ETHERNET network.

(45) The system **100**B may be implemented as a Long Term Evolution (LTE) radio access network providing wireless service using the LTE air interface. LTE is a standard developed by the 3GPP standards organization. In this configuration, the baseband controller **104** and RUs **106** together are used to implement an LTE Evolved Node B (also referred to here as an "eNodeB" or "eNB"). An eNB may be used to provide UEs **110** with mobile access to the wireless network operator's core network **112** to enable UEs **110** to wirelessly communicate data and voice (using, for example, Voice over LTE (VoLTE) technology). However, it should be noted that the present systems and methods may be used with other wireless protocols, e.g., the system **100**B may be implemented as a 3GPP 5G RAN providing wireless service using a 5G air interface, as described below.

(46) Also, in an exemplary LTE configuration, each core network **112** may be implemented as an Evolved Packet Core (EPC) **112** comprising standard LTE EPC network devices such as, for example, a mobility management entity (MME) and a Serving Gateway (SGW) and, optionally, a Home eNB gateway (HeNB GW) (not shown) and a Security Gateway (SeGW or SecGW) (not shown).

(47) Moreover, in an exemplary LTE configuration, each baseband controller **104** may communicate with the MME and SGW in the EPC core network **112** using the LTE S1 interface and communicates with eNBs using the LTE X2 interface. For example, the baseband controller **104** can communicate with an outdoor macro eNB (not shown) via the LTE X2 interface.

(48) Each baseband controller **104** and radio unit **106** can be implemented so as to use an air interface that supports one or more of frequency-division duplexing (FDD) and/or time-division duplexing (TDD). Also, the baseband controller **104** and the radio units **106** can be implemented to use an air interface that supports one or more of the multiple-input-multiple-output (MIMO), single-input-single-output (SISO), single-input-multiple-output (SIMO), and/or beam forming schemes. For example, the baseband controller **104** and the radio units **106** can implement one or more of the LTE transmission modes. Moreover, the baseband controller **104** and the radio units **106** can be configured to support multiple air interfaces and/or to support multiple wireless operators.

(49) In some configurations, in-phase, quadrature-phase (I/Q) data representing pre-processed baseband symbols for the air interface is communicated between the baseband controller **104** and

the RUs **106**. Communicating such baseband I/Q data typically requires a relatively high data rate front haul.

(50) In some configurations, a baseband signal can be pre-processed at a source RU **106** and converted to frequency domain signals (after removing guard band/cyclic prefix data, etc.) in order to effectively manage the front-haul rates, before being sent to the baseband controller **104**. Each RU **106** can further reduce the data rates by quantizing such frequency domain signals and reducing the number of bits used to carry such signals and sending the data. In a further simplification, certain symbol data/channel data may be fully processed in the source RU **106** itself and only the resultant information is passed to the baseband controller **104**.

(51) The Third Generation Partnership Project (3GPP) has adopted a layered model for the LTE radio access interface. Generally, some combination of the baseband controller **104** and RUs **106** perform analog radio frequency (RF) functions for the air interface as well as digital Layer 1 (L1), Layer 2 (L2), and Layer 3 (L3) (of the 3GPP-defined LTE radio access interface protocol) functions for the air interface. Any suitable split of L1-L3 processing (between the baseband controller **104** and RUs **106**) may be implemented. Where baseband signal I/Q data is front-hauled between the baseband controller **104** and the RUs **106**, each baseband controller **104** can be configured to perform all or some of the digital L1, L2, and L3 processing for the air interface. In this case, the L1 functions in each RU **106** is configured to implement all or some of the digital L1 processing for the air interface.

(52) Where a front-haul ETHERNET network **118** is not able to deliver the data rate need to front haul (uncompressed) I/Q data, the I/Q data can be compressed prior to being communicated over the ETHERNET network **118**, thereby reducing the data rate needed communicate such I/Q data over the ETHERNET network **118**.

(53) Data can be front-hauled between the baseband controller **104** and RUs **106** in other ways, for example, using front-haul interfaces and techniques specified in the Common Public Radio Interface (CPRI) and/or Open Base Station Architecture Initiative (OBSAI) family of specifications. The baseband controller **104** described herein may be similar to and/or perform at least some of the functionality of the O-RAN Distributed Unit (**0**-DU).

(54) Where functionality of a 5G DU **105** is discussed herein, it is equally applicable to a 5G CU **103** or a 4G baseband controller **104**. Therefore, where a C-RAN **100** is described herein, it may include 5G elements (as in FIG. **1**A) and/or 4G elements (as in FIG. **1**B).

(55) Example Multi-Operator System with Multi-Instance RUs

(56) FIG. **2**A is a block diagram illustrating a multi-operator **109** system **201**A with multi-instance RUs **106**A-B. It is understood that, even though a front-haul network **118** and a back-haul network **116** are not explicitly illustrated, they would generally be present in the system **201**A. Furthermore, the system **201**A may also have at least one core network **112** (e.g., one for each operator **109**), at least one management system **114** (e.g., one for each operator **109**), and possibly additional base stations nearby.

(57) Furthermore, it is understood that the designation of “network operator” or “operator” (and dotted line delineating such in FIG. **2**A) does not denote that all radio controllers **111** associated with a particular operator **109** are necessarily co-located. To the contrary, at least some of the radio controllers **111** associated with a particular operator **109** the gNB **100** may be physically remote from each other. It is understood that the specific configuration illustrated in FIG. **2**A is merely exemplary, and any number/type of each node may exist in the system **201**A.

(58) The system **201**A includes multiple network operators **109**A-B, such as T-Mobile®, Verizon®, or ATT®, each operator **109** deploying, configuring, and controlling various base stations, such as macro base station(s), C-RAN(s) **100**, and/or other small cell(s).

(59) The radio controllers **111**A-H may be CUs **103** or DUs **105** in 5G C-RAN(s) **100**A and/or baseband controllers **104** in 4G C-RAN(s) **100**B, e.g., any of which may operate according to O-RAN specifications. Each radio controller **111** may be assigned its own respective IP address (IP**1**-

IP**8** in FIG. **2**A). The radio controllers **111** associated with a particular operator **109** may belong to the same C-RAN **100** or different C-RANs **100**, e.g., a single operator **109** could implement multiple C-RANs **100**.

(60) Even though the radio controllers **111** correspond to different operators **109**, the radio controllers **111** may still be able to receive uplink signals, and to transmit downlink signals, using some of the same RUs **106**A-B. Each RU **106** may optionally implement more than one RU instance **107**A-H (e.g., modules). In some examples, each RU instance **107** within a multi-instance RU **106** may be implemented as an independent digital instance (e.g., a processing core) on one or more programmable processors in the multi-instance RU **106** within a single physical housing, e.g., where each programmable processor is a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or a digital signal processor (DSP).

(61) Each multi-instance RU **106**, like the ones shown in FIG. **2**A, may have multiple ETHERNET ports, each being assigned to a different RU instance **107**. The term "ETHERNET port" used herein generally refers, without limitation, to a logical port within an RU **106** within hardware on an RU **106**, e.g., a Transmission Control Protocol (TCP) port or a User Data gram Protocol (UDP) port. In some configurations, an RU **106** may have a single physical port and multiple logical ports (referred to as simply "ETHERNET ports"). Thus, each RU instance **107** may use a respective ETHERNET port. Thus, each RU instance **107** in an RU **106** can share the same physical port (and the same IP address) but have distinct communication sessions with external device(s).

(62) In FIG. **2**A, for example, in a first multi-instance RU **106**A, a first RU instance (C1) **107**A may be assigned to ETHERNET port 1 in the RU **106**A with IP address 1, a second RU instance (C2) **107**B may be assigned to ETHERNET port 2 in the RU **106**A with IP address 1, a third RU instance (C3) **107**C may be assigned to ETHERNET port 3 in the RU **106**A with IP address 1, and a fourth RU instance (C4) **107**D may be assigned to ETHERNET port 4 in the RU **106**A with IP address 1. Therefore, in some configurations, each RU instance **107** in a multi-instance RU **106** may use a different ETHERNET port, but all RU instances **107** in the multi-instance RU **106** may share the same IP address. Additionally, in some configurations, each RU instance **107** in a multi-instance RU **106** may share the same MAC address.

(63) As shown in FIG. **2**A, multiple RU instances **107** in a multi-instance RU **106** may each initiate a separate connection with the same radio controller **111** (on a separate ETHERNET port), e.g., each RU instance **107** being configured with a different NETCONF session. For example, the first RU instance (C1) **107**A and the second RU instance (C2) **107**B in the first RU **106** may each be assigned to (and be configured by) the first radio controller **111**A associated with the first operator **109**A. Similarly, each of the first three RU instances **107**E-G in the second RU **106** may each be assigned to (and be configured by) the first radio controller **111**A associated with the first operator **109**A.

(64) Alternatively or additionally, multiple RU instances **107** in a multi-instance RU **106** may each initiate a separate connection to different respective radio controllers **111** associated with the same operator **109**. For example, the second RU instance (C2) **107**B in the first RU **106** may be assigned to (and be configured by) the first radio controller **111**A associated with the first operator **109**A, while the third RU instance (C3) **107**C in the first RU **106** may be assigned to (and be configured by) the second radio controller **111**B associated with the first operator **109**A.

(65) Alternatively or additionally, multiple RU instances **107** in a multi-instance RU **106** may each initiate a separate connection to different radio controllers **111** associated with different operators **109**. For example, the third RU instance (C3) **107**C in the first RU **106** may be assigned to (and be configured by) the second radio controller **111**B associated with the first operator **109**A, while the fourth RU instance (C4) **107**B in the first RU **106** may each be assigned to (and be configured by) the first radio controller **111**E associated with the second operator **109**B.

(66) After an RU **106** powers up, it may be configured to discover the radio controller **111** it is assigned to and be configured by that radio controller **111** via a NETCONF session. Among other

things, the radio controller **111** may inform the RU's operational parameters, e.g., the RU's operating channel(s) (RF band), bandwidth, downlink and uplink carrier information, Root Sequence Index (RSI), Physical Cell Identity (PCI), operating power, user profile management information, file and log upload configuration, software management information, etc.).

(67) Additionally, the radio controller **111** may also indicate to the RU **106** how many carriers it will need to support for that radio controller **111**. In other words, the radio controller **111** may inform the RU **106**, during configuration, how many connections the RU **106** will have to maintain with the radio controller **111**, where a different connection from a different RU instance **107** may be established between the radio controller **111** and RU **106** for each carrier/connection. For example, the number of carriers may be communicated outside the NETCONF session, e.g., during transmission of DU IP addresses in response to DHCP Vendor Option data request from RU. In some examples, the radio controller **111** may signal the number of carriers via M-plane communication.

(68) Example Multi-gNB System with Multi-Instance RUs

(69) FIG. **2**B is a block diagram illustrating another multi-gNB system **201**B with multi-instance RUs **106**A-B. It is understood that the various nodes may be communicatively coupled to each other, even though not shown in FIG. **2**B. For example, the each DU **105**A-H would be communicatively coupled to at least one CU **103**A-H and each RU instance (C1-C8) **107**A-H. Therefore, even though a front-haul network **118** and a back-haul network **116** are not explicitly illustrated, they would generally be present. Furthermore, the system **201**B may also have at least one core network **112** (e.g., one for each operator **109** in the system **201**B), at least one management system **114** (e.g., one for each operator **109** in the system **201**B), and possibly additional base stations nearby.

(70) Furthermore, it is understood that the designation of “gNB” (and dotted line delineating such in FIG. **2**B) does not denote that all CUs **103** and DUs **105** serving the gNB **100** are necessarily co-located. To the contrary, at least some of the CUs **103** and/or DUs **105** serving the gNB **100** may be physically remote from each other.

(71) The system **201**B includes multiple gNBs **100**C-D, each corresponding to, and being controlled by, a particular network operator **109** such as T-Mobile®, Verizon®, or ATT®, each gNB **100** having at least one CU **103** and multiple DUs **105**. As noted briefly above, the CU **103** in each gNB **100** in FIG. **2**B may be split between a CU-CP **103**A, E that handles control plane functions and one or more CU-UPs **103**B-D, F-H that handle user plane functions. Accordingly, each gNB **100** in FIG. **2**B includes a single CU-CP **103**A, E and multiple CU-UPs **103**B-D, F-H. It is understood that the specific configuration illustrated in FIG. **2**B is merely exemplary, and any number/type of each node may exist in the system **201**B.

(72) As noted above, a DU **105**A-H may act as (and may be referred to herein as) a “radio controller” for one or more RUs **106** in a 5G C-RAN **100**A operating according to O-RAN specifications. Each DU **105** may communicate over its own respective IP address (DUIP1-DUIP8 in FIG. **2**B).

(73) Even though the gNBs **100** correspond to different operators **109**, both gNBs **100** may be configured to receive uplink signals, and to transmit downlink signals, using some of the same RU instances (C1-C8) **107**A-H. As before, each RU **106** may optionally implement more than one RU instance **107** (e.g., modules).

(74) As shown in FIG. **2**B, multiple RU instances **107** in a multi-instance RU **106** may initiate a separate connection with the same DU **105** (on a separate ETHERNET port), e.g., each RU instance **107** being configured with a different NETCONF session. For example, the first RU instance (C1) **107**A and the second RU instance (C2) **107**B in the first RU **106** may each be assigned to (and be configured by) a first DU **105**A in a first gNB **100**C. Similarly, each of the first three RU instances **107**E-G in the second RU **106** may each be assigned to (and be configured by) a first DU **105**A in a first gNB **100**C.

(75) Alternatively or additionally, multiple RU instances **107** in a multi-instance RU **106** may each initiate a separate connection to different respective DUs **105** in the same gNB **100**. For example, the second RU instance (C2) **107**B in the first RU **106** may be assigned to (and be configured by) the first DU **105**A in the first gNB **100**C, while the third RU instance (C3) **107**C in the first RU **106** may be assigned to (and be configured by) the second DU **105**B in the first gNB **100**C.
(76) Alternatively or additionally, multiple RU instances **107** in a multi-instance RU **106** may each initiate a separate connection to different DUs **105** in different respective gNBs **100**. For example, the third RU instance (C3) **107**C in the first RU **106** may be assigned to (and be configured by) the second DU **105**B in the first gNB **100**C, while the fourth RU instance (C4) **107**D in the first RU **106** may each be assigned to (and be configured by) the first DU **105**E in the second gNB **100**D.
(77) As will be discussed below, once an RU **106** powers up, it may be configured to discover the DU **105** it is assigned to and be configured by that DU **105** via a NETCONF session. Among other things, the DU **105** may inform the RU **106** how many carriers it will need to support for that DU **105**. In other words, the DU **105** may inform the RU **106**, during configuration, how many connections the RU **106** will have to maintain with the DU **105**, where a different connection from a different RU instance **107** may be established between the DU **105** and RU **106** for each carrier/connection. The DU **105** may signal the number of carriers via M-plane communication and start multiple NETCONF connections with the same RU **106** (if the DU **105** has multiple carriers configured for that RU **106**).
(78) First Example Configuration of Radio Controller Discovery in a System
(79) FIG. **3**A is a block diagram illustrating the first example configuration of radio controller **111** discovery in a system **301**. In the system **301** in FIG. **3**A, the gNB **100**E is implemented using one or more 5G C-RANs **100**, though other configurations are possible, e.g., C-RAN(s) **100** having 4G and/or 5G components. Thus, the radio controllers **111** in the system **301** of FIG. **3**A are DUs **105**A-D. Additionally, the gNB **100**E may include any number of CUs **103** or DUs **105** (or baseband controllers **104** in a 4G C-RAN **100**B). Additionally, the system **301** may include more than one gNB **100**E, e.g., in a multi-operator system where one or more gNBs **100** are associated with each of the network operators **109**.
(80) In the system **301**, a new RU **106**D is being powered up for the first time and needs to discover its radio controller(s) **111** (e.g., DU(s) **105** that will configure one or more RU instances **107** in the new RU **106**D). In the system **301**, an existing RU **106**C (that has already discovered its radio controller **111**) is illustrated, though there could be zero or more than one existing RU **106** at the time the first example configuration is performed, e.g., the new RU **106**D may be the first RU **106** powered up in a given C-RAN **100**. Furthermore, multiple new RUs **106**D may perform radio controller **111** discovery in overlapping, parallel, and/or sequential processes.
(81) The front-haul network **118** in first example configuration (and optionally the second example configuration discussed below) uses the Internet Protocol version 6 (IPv6) network protocol. IPv6 provides several different options to generate an IP address, two of which are discussed herein: a stateless address autoconfiguration mode (where a client on the IPv6 network derives its IP address from its Media Access Control (MAC) address) or a managed mode (where the client contacts a DHCP server for its IP address).
(82) FIG. **3**B is a block diagram illustrating different portions of an IPv6 address and how an IPv6 address can be derived from a MAC address **308**. An IPv6 address can be used in two different ways: as a link-local IP address (which a device can use to reach other devices in its link-local network, such as a front-haul network **118**, but cannot use to reach devices beyond its link-local network) and as a global IPv6 address (which a device can use to reach other devices beyond its link-local network). As such, a link-local IPv6 address **311** can include two different parts: a link-local IPv6 address prefix **304** and an Interface ID (IID) **306**, which is supposed to be unique on the links reached by routing to that prefix, thus giving an IPv6 address that is unique within the applicable scope (link local or global). Specifically, the link-local IPv6 address prefix **304** identifies

a network segment (link-local scope or global scope) and the Interface ID (IID) **306** uniquely identifies the host device. This combination of a link-local IPv6 address prefix **304** and an Interface ID (IID) **306** provides the full link-local IPv6 address **311**.

(83) To derive an Interface ID (IID) **306** of a device (e.g., an RU **106**), the device's MAC address **308** may be extended as follows: (1) 0xFFFE is inserted in between two halves of the device's MAC address **308** (0x001122 and 0xABCDEF in the FIG. **3**B example) to produce an EUI-64 Interface ID **310**; and (2) the 7th most significant bit (MSB) of the resulting EUI-64 Interface ID **310** is complemented/inverted to produce an Interface ID (IID) **306** for the device. The MAC address **308**, EUI-64 Interface ID **310**, and Interface ID (IID) **306** are illustrated in hexadecimal in FIG. **3**B. The MAC address **308** may be 48 bits long while the EUI-64 Interface ID **310** and Interface ID (IID) **306** are 64 bits long each.

(84) The first example configuration may utilize a stateless address autoconfiguration mode in which the new RU **106**D, upon powering up for the first time, initiates radio controller **111** discovery by: (1) determining its IP address (e.g., an IPv6 link-local IP address **311**) from its MAC address **308** (as illustrated in FIG. **3**B); and (2) sending a discovery message (e.g., discovery packet(s)) to a multicast address on the local link, e.g., a front-haul network **118** (shown in Step **2** in FIG. **3**A). For example, the multicast address may be an IPv6-protocol-specified multicast address, e.g., FF02:0:0:0:0:0:0:1 (the DU **105** should be on this all-node multicast address by default), or to a specific predetermined multicast address from the range FF02:0:0:0:0:1:FF00:0000-FF02:0:0:0:0:1:FFFF:FFF. In some examples, each RU **106** sends a single neighbor solicitation, while in other examples an RU **106** may send a respective neighbor solicitation for each RU instance **107** within the RU **106**.

(85) Networks using IPv6 addressing implement certain protocol messages for discovering neighboring devices. Neighbor Discovery Protocol is used for discovery devices on the link-local network. They are IPv6 Internet Control Message Protocol (ICMP) version 6 (ICMPv6) type packets and have different types like Router Solicitation (RS), Router Advertisement (RA), Neighbor Solicitation (NS), Neighbor Advertisement (NA), etc. In multicast messaging, the new RU **106**D sends messages to a multicast address, which is then sent to devices at other respective addresses. Using IPv6 Neighbor Discovery Protocol (NDP), a device can discover router(s) (using “router solicitation” and “router advertisement” messages) or other device(s) (using “neighbor solicitation” and “neighbor advertisement”), such as radio controllers **111**, in the link-local network. When a device (such as the new RU **106**D) powers up for the first time on an IPv6 network (such as a front-haul network **118**), it can send a multicast neighbor solicitation that is received by neighboring devices on the link-local network. Similarly, when a device comes up new on the IPv6 network, it can send a router solicitation and receive a router advertisement from a router on the link-local network. In some configurations, a header in the neighbor solicitation packet may include a value in a particular field (e.g., a value of 135 in a ICMPv6 packet header) that indicates that the packet includes a neighbor solicitation.

(86) There are at least two options for sending a neighbor solicitation message—directed (where the new RU **106**D sends the neighbor solicitation to a specific link-local address) or multicast (where the new RU **106**D sends the neighbor solicitation to all devices on the link-local network via a multicast address). By sending the neighbor solicitation to the multicast address, it will be sent to all radio controller(s) (e.g., DU(s) **105**A-D) on the link-local network (e.g., on the same front-haul network **118**), but only the radio controller **111** to which the new RU **106**D is assigned would initiate a NETCONF session in response. Thus, sending the neighbor solicitation by multicast may reduce the amount of retries and traffic in the event that a radio controller **111** doesn't know if the new RU **106**D is available and it does multiple retries before signaling that the new RU **106**D is not available and stops trying to reach the new RU **106**D. In a system **301** with 128 RUs **106** and multiple radio controllers **111**, the bandwidth savings can be substantial.

(87) In Step **3** of FIG. **3**A, the radio controllers **111** may listen for RU messages through either of

the following mechanism: neighbor solicitation (e.g., where the neighbor solicitation is sent to a default multicast address in the link-local network, such as FF02:0:0:0:0:1) and multicast messaging (e.g., where the neighbor solicitation is sent to a specific predetermined multicast address from the range FF02:0:0:0:0:1:FF00:0000-FF02:0:0:0:0:1:FFFF:FFF could be chosen as well to send the same message). In neighbor solicitation, the radio controller **111** may respond by directly initiating a NETCONF session (in Step **4** of FIG. **3**A), after which the RU **106** will cease sending neighbor solicitation messages. In multicast messaging, the radio controller **111** may respond to a multicast message (not a neighbor solicitation) from the new RU **106**D with a unicast message (not a neighbor advertisement) and/or directly initiate a NETCONF session, after which the new RU **106**D may cease sending multicast messages.
(88) The radio controllers **111** may determine an RU's IPv6 link-local address from the RU's discovery message (e.g., neighbor solicitation message). Additionally, each radio controller **111** may be configured with a whitelist of MAC addresses **308** for RUs **106** that have joined or may join the C-RAN **100** in the future and are assigned to that particular radio controller **111**. Accordingly, each radio controller can compare (1) the sending RU's IP address from the neighbor solicitation with (2) IP addresses derived from MAC addresses **308** in its respective whitelist.
(89) If the link-local IP address **311** (of the new RU **106**D that sent the neighbor solicitation message) corresponds to an IP address derived from a MAC address **308** in the radio controller's whitelist, the new RU **106**D is assigned to the radio controller **111**. Assume in FIG. **3**A that the first DU **105**A (acting as a radio controller **111**) determines that the MAC address **308** of the new RU **106**D is in its whitelist. In other words, at least one of the RU instances **107**E-H in the new RU **106**D is assigned to the first DU **105**A in FIG. **3**A.
(90) The new RU **106**D may then be authenticated and at least one of the RU instances **107**E-H in the new RU **106**D) configured by the radio controller(s) **111** it is assigned to (the first DU **105**A in this case) before the new RU **106**D can start providing service to UEs **110**. To do this, the radio controller **111** (the first DU **105**A) may initiate a configuration session with the new RU **106**D. NETCONF is a configuration management protocol that can be used by radio controllers **111** to configure at least one RU instance **107** in RUs **106** (or entire RUs **106** in the case of single-instance RUs **106**). NETCONF has its own protocols to initiate a session from either the server side or the client side. Once the NETCONF session(s) have been established, the radio controller **111** (the first DU **105**A in FIG. **3**A) may send configuration information to the new RU **106**D (in Step **5** of FIG. **3**A). In examples, an M-plane session is established between the radio controller **111** (the first DU **105**A in FIG. **3**A) and the new RU **106**D using the underlying NETCONF session, where the NETCONF session may use Transport Layer Security (TLS) or Secure Shell (SSH).
(91) Therefore, the first example configuration is more efficient than conventional radio controller **111** discovery in a distributed system because the new RU **106**D sends a neighbor solicitation and, in turn, receives a NETCONF session request from the radio controller **111**. Since the radio controller **111** (the first DU **105**A in FIG. **3**A) can quickly determine that the new RU **106**D sending the neighbor solicitation is one assigned to it (based on its whitelist and information from the neighbor solicitation), the radio controller **111** can initiate a NETCONF session to start configuring RU instance(s) in the new RU **106**D immediately.
(92) In contrast, the O-RAN proposed mechanism proposes only DHCP-based discovery of the Radio Controller by the RU **106**, where the RU **106** would get the radio controller's **111** IP address from the DHCP server and then initiate a NETCONF session by using call home.
(93) As described above, each RU **106** may have multiple (e.g., four) RU instances **107**, each able to connect to any radio controller **111**. Among other things, the radio controller **111** may inform the new RU **106**D of its operational parameters, e.g., operating channel(s), Root Sequence Index (RSI), Physical Cell Identity (PCI), operating power, etc.). The number of carriers may be communicated to the new RU **106**D outside the NETCONF session in some configurations, e.g., during transmission of DU IP addresses in response to DHCP Vendor Option data request from RU. In that

case, the new RU **106** would then initiate the respective number of NETCONF call homes (one for each RU instance **107**) or just a single call home and assign the desired number of RU instances **107** to operate with the specific radio controller **111**. In some configurations, a single NETFONF session may be used for all RU instances **107** connecting to the same radio controller **111**.

(94) Alternatively, a different NETCONF session may be used per RU instance **107** connecting to the same radio controller. Accordingly, in some configurations, the new RU **106**D may send information about its ports for the radio controller **111** to establish additional NETCONF sessions or trigger a NETCONF call home to the radio controller **111** to setup additional NETCONF sessions, one each for the number of carriers requested by the radio controller **111**. In some configurations where the new RU **106**D provides multiple-carrier service for a single radio controller **111**, a different respective RU instance **107** on the new RU **106**D may implement a different carrier. In examples, each RU instance **107** within the new RU **106**D may be implemented as an independent digital instance (e.g., a processing core) on one or more programmable processors in the new RU **106**D, e.g., where each programmable processor is a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or a digital signal processor (DSP).

(95) Method for Radio Controller Discovery According to First Example Configuration

(96) FIG. **4** is a flow diagram illustrating a method **400** for radio controller **111** discovery according to the first example configuration described herein. The method **400** may be performed in a system **301** with a C-RAN **100** (implementing 5G gNB(s) and/or 4G eNB(s)) and a new RU **106** joining the C-RAN **100**. The radio controller(s) **111** may be DUs **105** or CUs **103** (in 5G configurations) or baseband controllers **104** (in 4G configurations). In some configurations, the system **301** may include multiple gNBs **100** and/or eNBs **100** (e.g., each controlled by a different network operator **109**); multiple new RUs **106**D joining the C-RAN(s) **100** (each performing the method **400** sequentially and/or in parallel with other new RUs **106**D); and/or multiple existing RUs **106**C that have previously joined the C-RAN(s) **100**. Each of the devices in the system **301** may perform its respective steps using at least one respective processor that executes instructions stored in at least one respective memory.

(97) In some examples, the front-haul **118** of the C-RAN **100** utilizes Internet Protocol version 6 (IPv6) network protocol. The new RU **106**D (and other RUs **106** in the system **301**) may optionally implement more than one RU instance **107** (e.g., modules). In some configurations, different respective RU instances **107** on the new RU **106**D may implement different carriers. In examples, each RU instance **107** within the new RU **106**D may be implemented as an independent digital instance (e.g., a processing core) on one or more programmable processors in the new RU **106**D, e.g., where each programmable processor is a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or a digital signal processor (DSP).

(98) Additionally, the blocks of the flow diagram shown in FIG. **4** have been arranged in a generally sequential manner for ease of explanation; however, it is to be understood that this arrangement is merely exemplary, and it should be recognized that the processing associated with method **400** (and the blocks shown in FIG. **4**) can occur in a different order (for example, where at least some of the processing associated with the blocks is performed in parallel and/or in an event-driven manner). Also, most standard exception handling is not described for ease of explanation; however, it is to be understood that method **400** can and typically would include such exception handling.

(99) The method begins at optional step **402** where the new RU **106**D powers up for the first time in a link-local network (such as a front-haul network **118**). Generally, but without limitation, the new RU **106** would power up after being physically connected to a front-haul network **118** of a C-RAN **100**. Optional step **402** may be performed as part of initial deployment of one or more RUs **106** or following maintenance, upgrade, or repair of the new RU **106**D. Upon powering up, the new RU **106**D may have access to its MAC address **308**, but need to determine its own IP address and

discover radio controller(s) **111** it is assigned to (e.g., DU(s) **105** that will configure one or more RU instances **107** in the new RU **106**D).

(100) Accordingly, the method **400** proceeds at step **404** where the new RU **106**D determines its own IP address (e.g., a IPv6 link-local address within the front-haul network **118**) based on its MAC address **308**. For example, the new RU **106**D can autonomously determine its own IP address from its MAC address **308** (as described in FIG. 3B) without a need for a DHCP server.

(101) The method **400** proceeds at step **406** where the new RU **106**D sends a discovery message (e.g., discovery packet(s)) to all radio controllers **111** on the local link (e.g., a front-haul network **118**) via a multicast address. In some configurations, the discovery message(s) is/are IPv6 neighbor solicitation message(s), which are received by all radio controllers **111** on the same local link (e.g., the same front-haul network **118**) via the multicast address. In some examples, the multicast address may be pre-stored on the new RU **106**D. In some examples, the new RU **106**D may send (to the multicast address) a separate discovery message from each RU instance **107**E-H that it implements.

(102) The method proceeds at step **408** where each of the radio controllers **111** on the local link (e.g., the front-haul network **118**) receives the discovery message from the new RU **106**D, via the link-local network, and determines whether the new RU **106**D is assigned to the respective radio controller **111**. The radio controllers **111** may determine the RU's IPv6 link-local address from the new RU's discovery message (e.g., neighbor solicitation message). Additionally, each radio controller **111** may be configured with a whitelist of MAC addresses **308** for RUs **106** that have joined or may join the C-RAN **100** in the future and are assigned to that particular radio controller **111**. Accordingly, each radio controller can compare (1) the sending RU's IP address from the neighbor solicitation with (2) the IP addresses derived from MAC addresses **308** in its respective whitelist. If the link-local IP address **311** (of the new RU **106**D that sent the neighbor solicitation message) corresponds to an IP address derived from a MAC address **308** in the radio controller's whitelist, the new RU **106**D is assigned to the radio controller **111**.

(103) The method **400** proceeds at step **410** where each of at least one radio controller **111**, to which the new RU **106**D is assigned, establishes at least one configuration session with the new RU **106**D, e.g., only in response to determining that the link-local IP address (of the new RU **106**D that sent the neighbor solicitation message) corresponds to an IP address derived from a MAC address in the respective radio controller's whitelist. For example, the configuration session(s) may be NETCONF session(s).

(104) In some examples, the radio controller(s) **111** to which the new RU **106**D is assigned may respond to the new RU's **106**D neighbor solicitation by initiating a NETCONF session, after which the RU **106** will cease sending multicast neighbor solicitation messages. Alternatively, the radio controller(s) **111** to which the new RU **106**D is assigned may respond to the new RU's **106**D multicast message (not a neighbor solicitation) from the new RU **106**D by sending a unicast message (not a neighbor advertisement) and/or directly initiate a NETCONF session, after which the new RU **106**D will cease sending multicast neighbor solicitation messages. In examples, an M-plane session is established between the radio controller **111** and the new RU **106**D using the underlying NETCONF session, e.g., where the NETCONF session may use Transport Layer Security (TLS) or Secure Shell (SSH).

(105) Each radio controller **111**, to which the new RU **106**D is assigned, may then configure the new RU **106**D for operation via the respective at least one configuration session. This may include each radio controller **111**, to which the new RU **106**D is assigned, indicating to the new RU **106**D the number of carriers it needs to support for the respective radio controller **111**. For example, if the new RU **106**D has multiple (e.g., four) RU instances **107**, each may be able to connect to any radio controller **111**. Additionally, the new RU **106**D may send information about its ports for the radio controller **111** to establish additional NETCONF sessions or trigger a NETCONF call home to the radio controller **111** to setup additional NETCONF sessions, one each for the number of carriers

requested by the radio controller **111**. For example, different RU instances **107** in the new RU **106**D could connect to the same radio controller **111** or different radio controllers **111** (and optionally different radio controllers **111** belonging to different network operators **109**).

(106) Second Example Configuration of Radio Controller Discovery in a System

(107) FIG. **5** is a block diagram illustrating the second example configuration of radio controller **111** discovery in a system **501**. In the system **501** of FIG. **5**, at least two gNBs **100**F-G are implemented using one or more 5G C-RANs **100**, though other configurations are possible, e.g., C-RAN(s) **100** having 4G and/or 5G components. For example, each gNB **100**F-G may be operated by a different network operator **109**. Additionally, the gNBs **100**F-G may include any number of CUs **103** or DUs **105** (or baseband controllers **104** in a 4G C-RAN(s) **100**B).

(108) In the system **501**, a new RU **106**D is being powered up for the first time and needs to discover its radio controller(s) **111** (e.g., DU(s) **105** that will configure one or more RU instances **107**E-H in new RU **106**D). In the system **501**, an existing RU **106**C (that has already discovered its radio controller **111**) is illustrated, though there could be zero or more than one existing RU **106** at the time the first example configuration is performed, e.g., the new RU **106**D may be the first RU **106** powered up in a given C-RAN **100**. Furthermore, multiple new RUs **106**D may perform radio controller **111** discovery in parallel processes.

(109) As noted above, one O-RAN proposal is for the new RU **106**D to contact a Dynamic Host Configuration Protocol (DHCP) server **113** after power up, after which the DHCP server **113** assigns new RU **106**D an IP address and also sends the IP address of its radio controller **111** (DU **105** in FIG. **5**) to the new RU **106**D. There is at least one possible problem with this O-RAN proposal in systems **501** using multi-carrier RUs **106**. Specifically, this O-RAN proposal would limit communication to one radio controller **111** and one RU **106** because the DHCP server sends back to the RU **106** (in a vendor field) a single radio controller **111** IP address and a single RU **106** IP address. But if the new RU **106**D is capable of initiating connections with multiple radio controllers **111** (DUs **105** in FIG. **5**), the O-RAN proposal would effectively leave this functionality unused.

(110) Instead of the O-RAN proposal, the second example configuration described herein utilizes DHCP server(s) **113** that is/are capable of sending back IP addresses for multiple radio controllers **111** in the DHCP server's response to the new RU **106**D. If the radio controller **111** and new RU **106**D are both capable of initiating multiple connections to each other, the radio controller's **111** IP address may be listed twice in the DHCP server's response to the new RU **106**D, once for each carrier. Accordingly, when the new RU **106**D gets this response from the DHCP server, it knows how many connections to each radio controller **111** (DU in FIG. **5**) it needs to entertain/establish, one per carrier.

(111) As before, each radio controller **111** (DU **105** in FIG. **5**) may communicate over its own respective IP address (DUIP1-DUIP8 in FIG. **5**), while each RU instance **107**E-H in the new RU **106**D may use the same IP address (IP**2**) and MAC address **308** (MAC2) but different logical Ethernet ports (as shown in FIG. **5**). The DHCP server **113** may be configured (e.g., configured/updated during operation) with a list of RU MAC addresses **308** and the IP address(es) of the radio controller(s) **111** (DU(s) **105** in FIG. **5**) they are assigned to. In configurations with multiple DHCP servers **113** (not shown), each DHCP server **113** may have a range of IP addresses allocated to it, e.g., a first range of IP addresses allocated to the first DHCP server **113** may be non-overlapping with a second range of IP addresses allocated to the second DHCP server **113**.

(112) Upon powering up for the first time in the local link (e.g., a front-haul network **118**), the new RU **106**D may determine an IP address (e.g., an IPv6 link-local IP address **311**) for itself from its MAC address **308** (the MAC address **308** of the new RU **106**D would be MAC2 in FIG. **5**).

(113) In some configurations, the new RU(s) **106**D each perform a discovery process (e.g., after receiving a router advertisement (RA) on the link-local network). The discovery process can include the new RU(s) **106**D each sending a solicit message and receiving a DHCP server **113**

response, with information to connect to radio controller(s) **111**. The new RU **106**D can then receive the IP address(es) of the radio controller(s) **111** (DU(s) **105** in FIG. **5**) it needs to connect to the radio controller(s) **111**. For example, the new RU(s) **106**D can request Vendor Specific data (encapsulated in Option 17 message) from the DHCP server **113**, which will contain the DU IP addresses. The DHCP server **113** may validate the new RU(s) **106**D and then provide the new RU(s) **106**D with the Vendor Specific Data containing the DU IP addresses, in the Option 17 packet.

(114) The DHCP response will also indicate the number of carriers the radio controller **111** (DU **105** in FIG. **5**) needs the new RU **106**D to establish. For example, the new RU **106**D may receive, from the DHCP server **113**, a response indicating 'RU MAC2: DUIP1; DUIP1; DUIP1' (or 'RU MAC2: DUIP1:3') because the new RU **106**D needs to establish three different connections to the first DU **105**A in the first gNB **100**C (DUIP1 is listed three times to indicate three different carriers for the first DU **105**A in the first gNB **100**C). In contrast, when the existing RU **106**C previously performed radio controller discovery, it may have received (1) from the DHCP server **113**, a response indicating 'RU MAC1: DUIP1; DUIP1; DUIP2; DUIP5' (or 'RU MAC1: DUIP1:2; DUIP2; DUIP5') because the existing RU **106**C needs to establish two connections to the first DU **105**A, one connection to the second DU **105**B in the first gNB **100**C; and one connection to the first DU **105**E in the second gNB **100**D.

(115) Alternatively, in a system **501** with multiple DHCP servers **113** (not shown), an RU **106** that needs to make connections with radio controllers **111** in two different gNBs **100**C-D may receive a DHCP response from each multiple different DHCP servers **113**, e.g., where each DHCP server **113** serves a different gNB **100**C-D. For example, the new RU **106**D in FIG. **5** may receive, from the first DHCP server **113**A (that serves the first gNB **100**C), a response indicating 'RU MAC2: DUIP1; DUIP1; DUIP1' (or 'RU MAC2: DUIP1:3') because the new RU **106**D needs to establish three different connections to the first DU **105**A in the first gNB **100**C (DUIP1 is listed three times to indicate three different carriers for the first DU **105**A in the first gNB **100**C). In contrast, the existing RU **106**C performed radio controller discovery, it may have received (1) from the first DHCP server **113**A, a response indicating 'RU MAC1: DUIP1; DUIP1; DUIP2' (or 'RU MAC1: DUIP1:2; DUIP2') because the existing RU **106**C needs to establish two connections to the first DU **105**A and one connection to the second DU **105**B in the first gNB **100**C; and (2) from the second DHCP server **113**B (that serves the second gNB **100**D), a response indicating 'RU MAC1: DUIP5' because the existing RU **106**C needs to establish a connection to the first DU **105**E in the second gNB **100**D.

(116) The new RU **106**D may then initiate a different configuration session for each carrier it needs to establish with each radio controller **111** (DU **105** in FIG. **5**) it is assigned to. For example, the new RU **106**D may initiate three different NETCONF sessions the first DU **105**A in the first gNB **100**C (using DUIP1), one NETCONF session for each of the carriers the new RU **106**D needs to establish. Once the NETCONF session has been established, the first DU **105**A may send configuration information to the new RU **106**D. In examples, an M-plane session is established between the radio controller **111** and the new RU **106**D using the underlying NETCONF session, e.g., where the NETCONF session may use Transport Layer Security (TLS) or Secure Shell (SSH).

(117) As noted earlier, the O-RAN proposal does not account for how a multi-carrier RP in a distributed RAN (ORAN framework) could be deployed in multi-operator cases. Accordingly, the second example configuration of radio controller discover enables multi-carrier RUs **106** in multi-operator systems **501**.

(118) Method for Radio Controller Discovery According to Second Example Configuration

(119) FIG. **6** is a flow diagram illustrating a method **600** for radio controller **111** discovery according to the second example configuration described herein. The method **600** may be performed in a system **501** with a C-RAN **100** (implementing 5G gNB(s) and/or 4G eNB(s)) and a new RU **106** joining the C-RAN **100**. The radio controller(s) **111** may be DUs **105** or CUs **103** (in

5G configurations) or baseband controllers **104** (in 4G configurations). In some configurations, the system **501** may include multiple gNBs **100** and/or eNBs **100** (e.g., where each is controlled by a different network operator **109**); multiple new RUs **106**D joining the C-RAN(s) **100** (each performing the method **600** sequentially and/or in parallel with other new RUs **106**D); and/or less than or more than one existing RU **106**C that has/have previously joined the C-RAN(s) **100**. Each of the devices in the system **501** may perform its respective steps using at least one respective processor that executes instructions stored in at least one respective memory. In some examples, the front-haul **118** of each C-RAN **100** (e.g., gNB **100**C-D or eNB) utilizes Internet Protocol version 6 (IPv6) network protocol.
(120) The system **501** may also include at least one Dynamic Host Configuration Protocol (DHCP) server **113**, each of which are configured with a list of MAC addresses **308** for RU(s) **106** in the system and a list of IP address(es) of the radio controller(s) **111** each of the RU(s) **106** is assigned to.
(121) The new RU **106**D (and any existing RUs **106**C in the system **501**) may optionally implement more than one RU instance **107** (e.g., modules). In some configurations, different respective RU instances **107** on the new RU **106**D may implement different carriers. For example: (1) multiple RU instances **107** may support the same carrier; (2) one RU instance **107** may support multiple carriers; and/or (3) one RU instance **107** (e.g., in a multi-instance RU **106**) supports multiple carriers.
(122) In some cases, the new RU **106**D may establish multiple carriers for a single radio controller **111** to which it is assigned. In some configurations, each RU instance **107** within the new RU **106**D may be implemented as an independent digital instance (e.g., a processing core) on one or more programmable processors in the new RU **106**D, e.g., where each programmable processor is a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or a digital signal processor (DSP).
(123) Each radio controller **111** (DU **105** in the system **501**) may communicate over its own respective IP address (DUIP1-DUIP8 in the system **501**), while each RU instance **107**E-H in the new RU **106**D may use the same IP address (IP**2**) and MAC address **308** (MAC2) but different logical Ethernet port as shown in the system **501**. Additionally, the DHCP server **113** (in the same link-local network, e.g., fronthaul network **118**) may have a range of IP addresses allocated to it, e.g., a first range of IP addresses allocated to the first DHCP server **113**A may be non-overlapping with a second range of IP addresses allocated to the second DHCP server **113**B.
(124) Additionally, the blocks of the flow diagram shown in FIG. **6** have been arranged in a generally sequential manner for ease of explanation; however, it is to be understood that this arrangement is merely exemplary, and it should be recognized that the processing associated with method **600** (and the blocks shown in FIG. **6**) can occur in a different order (for example, where at least some of the processing associated with the blocks is performed in parallel and/or in an event-driven manner). Also, most standard exception handling is not described for ease of explanation; however, it is to be understood that method **600** can and typically would include such exception handling.
(125) The method begins at step **602** where the new RU **106**D powers up for the first time in a link-local network (such as a front-haul network **118**, for example). Generally, but without limitation, the new RU **106** would power up after (or shortly before) being physically connected to a front-haul network **118** of a C-RAN **100**. Step **602** may be performed as part of initial deployment of one or more RUs **106** or following maintenance, upgrade, or repair of the new RU **106**D. Upon powering up, the new RU **106**D may have access to its MAC address **308**, but needs to determine its own IP address and discover radio controller(s) **111** it is assigned to (e.g., DU(s) **105** that will configure one or more RU instances **107** in the new RU **106**D).
(126) The method **600** proceeds at step **604** where the new RU **106**D requests, from the DHCP server **113**, IP address(es) for each radio controller **111** (e.g., DU **105**) it is assigned to. In some

configurations, the new RU **106**D performs a discovery process to establish a connection to the DHCP server **113**, as described above. Examples herein describe the DHCP server **113** operating in a link-local IP address environment, though the DHCP server **113** could alternatively send a global IP address of the radio controller(s) **111**. Alternatively, the new RU(s) **106**D are pre-configured with radio controller's IP address (e.g., an IPv6 link-local IP address **311**).
(127) The method **600** proceeds at step **606** where the DHCP server **113** sends a response to the new RU **106**D indicating a number of IP addresses equal to a number of carriers the new RU **106**D needs to establish, each of the IP addresses belonging to a radio controller **111** the new RU **106**D is assigned to.
(128) The DHCP response may list/indicate the new RU's MAC address **308** and one or more radio controller IP addresses that the new RU **106**D needs to initiate Netconf connection with. If the new RU **106**D is to establish N (N>=1) carriers for a particular radio controller **111**, the particular radio controller's IP address may be listed N times, e.g., a DHCP response similar to ‘RU MAC2: DUIP1; DUIP1; DUIP1’ (or ‘RU MAC2: DUIP1:3’) may indicate that the new RU **106**D (with a MAC address **308** of MAC2) needs to establish three different connections to the radio controller **111** with an IP address of DUIP1 (the first DU **105**A in the first gNB **100**C in the system **501**).
(129) In some scenarios, an RU **106** (such as the existing RU **106**C) needs to establish carriers for radio controllers **111** associated with different network operators. During radio controller discovery, the existing RU **106**C can request and receive Vendor Specific data (encapsulated in Option 17 message) from the DHCP server **113**. For example, the existing RU **106**C may receive a single response from a single DHCP server **113** indicating the number of carriers and associated IP addresses for the associated radio controllers **111** even though different radio controllers **111** in the list may be associated with different network operators **109**, e.g., the existing RU **106**C in the system **501** may receive a single DHCP response of ‘RU MAC1: DUIP1; DUIP1; DUIP2; DUIP5’, ‘RU MAC1: DUIP1:2; DUIP2; DUIP5’, or similar.
(130) Alternatively, the new RU **106**D may receive separate DHCP responses, each associated with a network operator that controls at least one radio controller **111** to which the new RU **106**D is assigned. For example, with reference to FIG. **5**, the existing RU **106**C would receive (during radio controller **111** discovery): (1) a first DHCP response indicating ‘RU MAC1: DUIP1; DUIP1; DUIP2’ (because the existing RU **106**C needed to establish two connections to the first DU **105**A and one connection to the second DU **105**B in the first gNB **100**C; and (2) a second DHCP response indicating ‘RU MAC1: DUIP5’ because the existing RU **106**C needed to establish a connection to the first DU **105**E in the second gNB **100**D.
(131) In some configurations, the DHCP server **113** also sends the new RU's IP address (e.g., IPv6 address) in the response. For example, the DHCP server **113** can allocate an IP address for the new RU **106**D only in response to successfully validating the new RU's MAC address. Alternatively, the new RU **106**D may determine its own IP address from its MAC address, e.g., as described in FIG. **3**B.
(132) The method **600** proceeds to step **608** where the new RU **106**D initiates a different configuration session for each carrier it needs to establish with each radio controller **111** it is assigned to. For example, the new RU **106**D may initiate multiple different NETCONF sessions with a particular radio controller **111**, one NETCONF session for each of the carriers the new RU **105**D needs to establish with the particular radio controller **111**. Once the NETCONF session has been established with a radio controller **111**, the radio controller **111** may send configuration information to the new RU **106**D. In examples, an M-plane session is established between the radio controller **111** and the new RU **106**D using the underlying NETCONF session, e.g., where the NETCONF session may use Transport Layer Security (TLS) or Secure Shell (SSH).
(133) The methods and techniques described here may be implemented in digital electronic circuitry, or with a programmable processor (for example, a special-purpose processor or a general-purpose processor such as a computer) firmware, software, or in combinations of them. Apparatus

embodying these techniques may include appropriate input and output devices, a programmable processor, and a storage medium tangibly embodying program instructions for execution by the programmable processor. A process embodying these techniques may be performed by a programmable processor executing a program of instructions to perform desired functions by operating on input data and generating appropriate output. The techniques may advantageously be implemented in one or more programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device. Generally, a processor will receive instructions and data from a read-only memory and/or a random access memory. For example, where a computing device is described as performing an action, the computing device may carry out this action using at least one processor executing instructions stored on at least one memory. Storage devices suitable for tangibly embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, such as EPROM, EEPROM, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and DVD disks. Any of the foregoing may be supplemented by, or incorporated in, specially-designed application-specific integrated circuits (ASICs).
(134) Terminology
(135) Brief definitions of terms, abbreviations, and phrases used throughout this application are given below.
(136) The term “determining” and its variants may include calculating, extracting, generating, computing, processing, deriving, modeling, investigating, looking up (e.g., looking up in a table, a database, or another data structure), ascertaining and the like. Also, “determining” may also include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.
(137) The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on”. Additionally, the term “and/or” means “and” or “or”. For example, “A and/or B” can mean “A”, “B”, or “A and B”. Additionally, “A, B, and/or C” can mean “A alone,” “B alone,” “C alone,” “A and B,” “A and C,” “B and C” or “A, B, and C.”
(138) The terms “connected”, “coupled”, and “communicatively coupled” and related terms may refer to direct or indirect connections. If the specification states a component or feature “may,” “can,” “could,” or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.
(139) The terms “responsive” or “in response to” may indicate that an action is performed completely or partially in response to another action. The term “module” refers to a functional component implemented in software, hardware, or firmware (or any combination thereof) component.
(140) The methods disclosed herein comprise one or more steps or actions for achieving the described method. Unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.
(141) In conclusion, the present disclosure provides novel systems, methods, and arrangements for discovering a radio controller in a C-RAN. While detailed descriptions of one or more configurations of the disclosure have been given above, various alternatives, modifications, and equivalents will be apparent to those skilled in the art without varying from the spirit of the disclosure. For example, while the configurations described above refer to particular features, functions, procedures, components, elements, and/or structures, the scope of this disclosure also includes configurations having different combinations of features, functions, procedures, components, elements, and/or structures, and configurations that do not include all of the described

features, functions, procedures, components, elements, and/or structures. Accordingly, the scope of the present disclosure is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof. Therefore, the above description should not be taken as limiting.

EXAMPLES

(142) Example 1 includes a distributed radio access network (RAN), comprising: a new radio unit (RU) being connected to a front-haul network and configured to: determine its own Internet Protocol (IP) address based on its Media Access Control (MAC) address; send a discovery message to all radio controllers in the front-haul network via a multicast address; and a plurality of radio controllers communicatively coupled to the new RU via the front-haul network, wherein each of the plurality of radio controllers are configured to receive the discovery message and determine whether the new RU is assigned to the respective radio controller; wherein each of at least one radio controller, to which the new RU is assigned, establishes at least one configuration session with the new RU.

(143) Example 2 includes the distributed RAN of Example 1, wherein the discovery message is a multicast Internet Protocol version 6 (IPv6) neighbor solicitation message.

(144) Example 3 includes the distributed RAN of any of Examples 1-2, wherein each of the plurality of radio controllers is configured to determine whether the new RU is assigned to the respective radio controller by determining whether a link-local IP address of the new RU corresponds to an IP address derived from a MAC address in a whitelist stored in the respective radio controller.

(145) Example 4 includes the distributed RAN of any of Examples 1-3, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.

(146) Example 5 includes the distributed RAN of any of Examples 1-4, wherein each of the at least one radio controller, to which the new RU is assigned, uses at least one NETCONF session to indicate a number of carriers that the new RU needs to establish for the respective radio controller.

(147) Example 6 includes the distributed RAN of any of Examples 1-5, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.

(148) Example 7 includes the distributed RAN of Example 6, wherein each of the multiple RU instances establishes a different carrier for a same one of the at least one radio controller to which the new RU is assigned.

(149) Example 8 includes the distributed RAN of any of Examples 6-7, wherein more than one of the multiple RU instances establishes a same carrier for a same one of the at least one radio controller to which the new RU is assigned.

(150) Example 9 includes the distributed RAN of any of Examples 6-8, wherein one RU instance in a multiple-instance RU establishes multiple carriers for a same one of the at least one radio controller to which the new RU is assigned.

(151) Example 10 includes the distributed RAN of any of Examples 1-9, wherein the radio controllers are implemented in Third Generation Partnership Project Fifth Generation (5G) Central Units or Distributed Units, Third Generation Partnership Project Long Term Evolution (LTE) baseband controllers, or a head unit in a Distributed Antenna System (DAS).

(152) Example 11 includes a method for radio controller discovery performed in a distributed radio access network (RAN), comprising: determining, at a new RU, its own Internet Protocol (IP) address based on its Media Access Control (MAC) address; sending, from the new RU, a discovery message to all radio controllers in a front-haul network via a multicast address; receiving, at each of a plurality of radio controllers communicatively coupled to the new RU via the front-haul network, the discovery message and determining whether the new RU is assigned to the respective

radio controller; and establishing at least one configuration session with the new RU.
(153) Example 12 includes the method of Example 11, wherein the discovery message is a multicast Internet Protocol version 6 (IPv6) neighbor solicitation message.
(154) Example 13 includes the method of any of Examples 11-12, further comprising determining, at each of the plurality of radio controllers, whether the new RU is assigned to the respective radio controller by determining whether a link-local IP address of the new RU corresponds to an IP address derived from a MAC address in a whitelist stored in the respective radio controller.
(155) Example 14 includes the method of any of Examples 11-13, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.
(156) Example 15 includes the method of any of Examples 11-14, wherein each of the at least one radio controller, to which the new RU is assigned, uses at least one NETCONF session to indicate a number of carriers that the new RU needs to establish for the respective radio controller.
(157) Example 16 includes the method of any of Examples 11-15, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.
(158) Example 17 includes the method of Example 16, wherein each of the multiple RU instances establishes a different carrier for a same one of the at least one radio controller to which the new RU is assigned.
(159) Example 18 includes the method of any of Examples 16-17, wherein more than one of the multiple RU instances establishes a same carrier for a same one of the at least one radio controller to which the new RU is assigned.
(160) Example 19 includes the method of any of Examples 16-18, wherein one RU instance in a multiple-instance RU establishes multiple carriers for a same one of the at least one radio controller to which the new RU is assigned.
(161) Example 20 includes the method of any of Examples 11-19, wherein the radio controllers are implemented in Third Generation Partnership Project Fifth Generation (5G) Central Units or
(162) Distributed Units, Third Generation Partnership Project Long Term Evolution (LTE) baseband controllers, or a head unit in a Distributed Antenna System (DAS).
(163) Example 21 includes a distributed radio access network (RAN), comprising: a new radio unit (RU) being connected to a plurality of radio controllers via a front-haul network and configured to request, from a Dynamic Host Configuration Protocol (DHCP) server after the new RU powers up a first time, IP addresses for each radio controller it is assigned to; and the DHCP server configured to send a response to the new RU indicating a number of Internet Protocol (IP) addresses equal to a number of carriers the new RU needs to establish, each of the IP addresses belonging to a radio controller the new RU is assigned to; wherein the new RU initiates a different configuration session for each carrier it needs to establish with each radio controller it is assigned to.
(164) Example 22 includes the distributed RAN of Example 21, wherein the response from the DHCP server also indicates the new RU's Internet Protocol (IP) address.
(165) Example 23 includes the distributed RAN of any of Examples 21-22, wherein the new RU determines its own Internet Protocol (IP) address based on its Media Access Control (MAC) address.
(166) Example 24 includes the distributed RAN of any of Examples 21-23, wherein the IP addresses are Internet Protocol version 6 (IPv6), Internet Protocol version 4 (IPv4) addresses, or a combination of both.
(167) Example 25 includes the distributed RAN of any of Examples 21-24, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.
(168) Example 26 includes the distributed RAN of any of Examples 21-25, wherein the response

from the DHCP server indicates a number of carriers that the new RU needs to establish with each radio controller to which it is assigned.

(169) Example 27 includes the distributed RAN of any of Examples 21-26, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.

(170) Example 28 includes the distributed RAN of Example 27, further comprising establishing, at each of the multiple RU instances, a different carrier for a same radio controller to which the new RU is assigned.

(171) Example 29 includes the distributed RAN of any of Examples 27-28, wherein each RU instance is configured using a separate configuration session.

(172) Example 30 includes the distributed RAN of any of Examples 28-29, wherein the response indicates IP addresses for radio controllers associated with different network operators.

(173) Example 31 includes a method for radio controller discovery performed in a distributed radio access network (RAN), comprising: requesting, from a Dynamic Host Configuration Protocol (DHCP) server after a new radio unit (RU) powers up a first time in a front-haul network, IP addresses for each radio controller the new RU is assigned to; sending, from the DHCP server, a response to the new RU indicating a number of Internet Protocol (IP) addresses equal to a number of carriers the new RU needs to establish, each of the IP addresses belonging to a radio controller the new RU is assigned to; and initiating, by the new RU, a different configuration session for each carrier it needs to establish with each radio controller it is assigned to.

(174) Example 32 includes the method of Example 31, wherein the response from the DHCP server also indicates the new RU's Internet Protocol (IP) address.

(175) Example 33 includes the method of any of Examples 31-32, wherein the new RU determines its own Internet Protocol (IP) address based on its Media Access Control (MAC) address.

(176) Example 34 includes the method of any of Examples 31-33, wherein the IP addresses are Internet Protocol version 6 (IPv6), Internet Protocol version 4 (IPv4) addresses, or a combination of both.

(177) Example 35 includes the method of any of Examples 31-34, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.

(178) Example 36 includes the method of any of Examples 31-35, wherein the response from the DHCP server indicates a number of carriers that the new RU needs to establish with each radio controller to which it is assigned.

(179) Example 37 includes the method of any of Examples 31-36, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.

(180) Example 38 includes the method of Example 37, further comprising establishing, at each of the multiple RU instances, a different carrier for a same radio controller to which the new RU is assigned.

(181) Example 39 includes the method of any of Examples 37-38, further comprising configuring each RU instance using a separate configuration session.

(182) Example 40 includes the method of any of Examples 31-39, wherein the response indicates IP addresses for radio controllers associated with different network operators.

Claims

1. A distributed radio access network (RAN), comprising: a new radio unit (RU) being connected to a front-haul network and configured to: determine its own Internet Protocol (IP) address based on

its Media Access Control (MAC) address; send a discovery message to all radio controllers in the front-haul network via a multicast address; and a plurality of radio controllers communicatively coupled to the new RU via the front-haul network, wherein each of the plurality of radio controllers are configured to receive the discovery message and determine whether the new RU is assigned to the respective radio controller; wherein each of at least one radio controller, to which the new RU is assigned, establishes at least one configuration session with the new RU.
2. The distributed RAN of claim 1, wherein the discovery message is a multicast Internet Protocol version 6 (IPv6) neighbor solicitation message.
3. The distributed RAN of claim 1, wherein each of the plurality of radio controllers is configured to determine whether the new RU is assigned to the respective radio controller by determining whether a link-local IP address of the new RU corresponds to an IP address derived from a MAC address in a whitelist stored in the respective radio controller.
4. The distributed RAN of claim 1, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.
5. The distributed RAN of claim 1, wherein each of the at least one radio controller, to which the new RU is assigned, uses at least one NETCONF session to indicate a number of carriers that the new RU needs to establish for the respective radio controller.
6. The distributed RAN of claim 1, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.
7. The distributed RAN of claim 6, wherein each of the multiple RU instances establishes a different carrier for a same one of the at least one radio controller to which the new RU is assigned.
8. The distributed RAN of claim 6, wherein more than one of the multiple RU instances establishes a same carrier for a same one of the at least one radio controller to which the new RU is assigned.
9. The distributed RAN of claim 6, wherein one RU instance in a multiple-instance RU establishes multiple carriers for a same one of the at least one radio controller to which the new RU is assigned.
10. The distributed RAN of claim 1, wherein the radio controllers are implemented in Third Generation Partnership Project Fifth Generation (5G) Central Units or Distributed Units, Third Generation Partnership Project Long Term Evolution (LTE) baseband controllers, or a head unit in a Distributed Antenna System (DAS).
11. A method for radio controller discovery performed in a distributed radio access network (RAN), comprising: determining, at a new RU, its own Internet Protocol (IP) address based on its Media Access Control (MAC) address; sending, from the new RU, a discovery message to all radio controllers in a front-haul network via a multicast address; receiving, at each of a plurality of radio controllers communicatively coupled to the new RU via the front-haul network, the discovery message and determining whether the new RU is assigned to the respective radio controller; and establishing at least one configuration session with the new RU.
12. The method of claim 11, wherein the discovery message is a multicast Internet Protocol version 6 (IPv6) neighbor solicitation message.
13. The method of claim 11, further comprising determining, at each of the plurality of radio controllers, whether the new RU is assigned to the respective radio controller by determining whether a link-local IP address of the new RU corresponds to an IP address derived from a MAC address in a whitelist stored in the respective radio controller.
14. The method of claim 11, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.
15. The method of claim 11, wherein each of the at least one radio controller, to which the new RU is assigned, uses at least one NETCONF session to indicate a number of carriers that the new RU needs to establish for the respective radio controller.

16. The method of claim 11, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.
17. The method of claim 16, wherein each of the multiple RU instances establishes a different carrier for a same one of the at least one radio controller to which the new RU is assigned.
18. The method of claim 16, wherein more than one of the multiple RU instances establishes a same carrier for a same one of the at least one radio controller to which the new RU is assigned.
19. The method of claim 16, wherein one RU instance in a multiple-instance RU establishes multiple carriers for a same one of the at least one radio controller to which the new RU is assigned.
20. The method of claim 11, wherein the radio controllers are implemented in Third Generation Partnership Project Fifth Generation (5G) Central Units or Distributed Units, Third Generation Partnership Project Long Term Evolution (LTE) baseband controllers, or a head unit in a Distributed Antenna System (DAS).
21. A distributed radio access network (RAN), comprising: a new radio unit (RU) being connected to a plurality of radio controllers via a front-haul network and configured to request, from a Dynamic Host Configuration Protocol (DHCP) server after the new RU powers up a first time, IP addresses for each radio controller it is assigned to; and the DHCP server configured to send a response to the new RU indicating a number of Internet Protocol (IP) addresses equal to a number of carriers the new RU needs to establish, each of the IP addresses belonging to a radio controller the new RU is assigned to; wherein the new RU initiates a different configuration session for each carrier it needs to establish with each radio controller it is assigned to.
22. The distributed RAN of claim 21, wherein the response from the DHCP server also indicates the new RU's Internet Protocol (IP) address.
23. The distributed RAN of claim 21, wherein the new RU determines its own Internet Protocol (IP) address based on its Media Access Control (MAC) address.
24. The distributed RAN of claim 21, wherein the IP addresses are Internet Protocol version 6 (IPv6), Internet Protocol version 4 (IPv4) addresses, or a combination of both.
25. The distributed RAN of claim 21, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.
26. The distributed RAN of claim 21, wherein the response from the DHCP server indicates a number of carriers that the new RU needs to establish with each radio controller to which it is assigned.
27. The distributed RAN of claim 21, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.
28. The distributed RAN of claim 27, further comprising establishing, at each of the multiple RU instances, a different carrier for a same radio controller to which the new RU is assigned.
29. The distributed RAN of claim 27, wherein each RU instance is configured using a separate configuration session.
30. The distributed RAN of claim 28, wherein the response indicates IP addresses for radio controllers associated with different network operators.
31. A method for radio controller discovery performed in a distributed radio access network (RAN), comprising: requesting, from a Dynamic Host Configuration Protocol (DHCP) server after a new radio unit (RU) powers up a first time in a front-haul network, IP addresses for each radio controller the new RU is assigned to; sending, from the DHCP server, a response to the new RU indicating a number of Internet Protocol (IP) addresses equal to a number of carriers the new RU

needs to establish, each of the IP addresses belonging to a radio controller the new RU is assigned to; and initiating, by the new RU, a different configuration session for each carrier it needs to establish with each radio controller it is assigned to.

32. The method of claim 31, wherein the response from the DHCP server also indicates the new RU's Internet Protocol (IP) address.

33. The method of claim 31, wherein the new RU determines its own Internet Protocol (IP) address based on its Media Access Control (MAC) address.

34. The method of claim 31, wherein the IP addresses are Internet Protocol version 6 (IPv6), Internet Protocol version 4 (IPv4) addresses, or a combination of both.

35. The method of claim 31, wherein each configuration session is a Network Configuration Protocol (NETCONF) session that configures the new RU to be operational in the distributed RAN.

36. The method of claim 31, wherein the response from the DHCP server indicates a number of carriers that the new RU needs to establish with each radio controller to which it is assigned.

37. The method of claim 31, wherein the new RU implements multiple RU instances in a single physical housing; wherein each RU instance in the new RU utilizes a separate logical Ethernet port on the new RU; and wherein each RU instance in the new RU shares a same physical Ethernet port and Media Access Control (MAC) address.

38. The method of claim 37, further comprising establishing, at each of the multiple RU instances, a different carrier for a same radio controller to which the new RU is assigned.

39. The method of claim 37, further comprising configuring each RU instance using a separate configuration session.

40. The method of claim 31, wherein the response indicates IP addresses for radio controllers associated with different network operators.